

WEEKLY PROGRESS REPORT – WEEK 04

Student Information

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Department: B.Tech Information Technology

Project Title: Civic Website / Geo-Location Based Systems Development

Reporting Week: Week 04

Overview

This week focused on integrating AI-assisted complaint support using the Google Gemini API, improving complaint booking workflows, refining the user interface, optimizing frontend state management, and fixing bugs identified during testing. Additional effort was spent improving responsiveness and preparing the application for demonstration.

Tasks Completed

- Integrated Google Gemini API to provide AI assistance within the Civic Website.
- Improved complaint registration and booking workflow.
- Enhanced React frontend components and overall UI/UX responsiveness.
- Refactored API handling and optimized state management.
- Fixed authentication, routing, and validation issues.
- Performed module testing and resolved reported bugs.

Milestones Achieved

- Successfully integrated AI-powered assistance.
- Complaint workflow became more reliable.
- Improved application responsiveness across devices.
- Increased overall system stability through debugging.

Challenges and Hurdles Faced

Challenge 1: Integrating Gemini API with the existing application while maintaining smooth response handling.

Solution: Implemented proper API request handling, response parsing, loading indicators, and error handling.

Challenge 2: Maintaining consistent application state across multiple React components.

Solution: Optimized state updates, reduced unnecessary re-renders, and improved component communication.

Lessons Learned

- Gained practical experience integrating generative AI into a web application.
- Improved understanding of API integration, asynchronous programming, and frontend optimization.
- Enhanced debugging, testing, and problem-solving skills.

Future Plan

- Enhance AI features with smarter complaint recommendations.
- Complete remaining backend improvements.
- Optimize database operations and prepare the project for final deployment and presentation.

Conclusion

Week 04 significantly enhanced the Civic Website by introducing AI-assisted functionality, improving workflow efficiency, refining the user experience, and increasing application stability. The project is now closer to a production-ready state.